The Complete ASCII Reference Table

Decimal, Hexadecimal, Binary, and Characters

ASCII (American Standard Code for Information Interchange) is a character encoding standard for electronic communication. It comprises 128 characters (0-127). The first 32 characters (and the last one, 127) are non-printing control characters used to manage hardware or data streams. Characters 32-126 are printable symbols.

1. Control Characters (0-31 & 127)

Dec	Hex	Binary	Symbol / Description	Dec	Hex	Binary	Symbol / Description
0	00	00000000	NUL (Null)	16	10	00010000	DLE (Data Link Escape)
1	01	00000001	SOH (Start of Heading)	17	11	00010001	DC1 (Device Ctrl 1)
2	02	00000010	STX (Start of Text)	18	12	00010010	DC2 (Device Ctrl 2)
3	03	00000011	ETX (End of Text)	19	13	00010011	DC3 (Device Ctrl 3)
4	04	00000100	EOT (End of Trans.)	20	14	00010100	DC4 (Device Ctrl 4)
5	05	00000101	ENQ (Enquiry)	21	15	00010101	NAK (Neg. Acknowledge)
6	06	00000110	ACK (Acknowledge)	22	16	00010110	SYN (Sync. Idle)
7	07	00000111	BEL (Bell)	23	17	00010111	ETB (End Trans. Block)
8	08	00001000	BS (Backspace)	24	18	00011000	CAN (Cancel)
9	09	00001001	HT (Horiz. Tab)	25	19	00011001	EM (End of Medium)
10	0A	00001010	LF (Line Feed)	26	1A	00011010	SUB (Substitute)
11	0B	00001011	VT (Vert. Tab)	27	1B	00011011	ESC (Escape)
12	0C	00001100	FF (Form Feed)	28	1C	00011100	FS (File Separator)
13	0D	00001101	CR (Carriage Return)	29	1D	00011101	GS (Group Separator)
14	0E	00001110	SO (Shift Out)	30	1E	00011110	RS (Record Separator)
15	0F	00001111	SI (Shift In)	31	1F	00011111	US (Unit Separator)
127	7F	01111111	DEL (Delete)				

2. Printable Characters (32-126)

Dec	Hex	Bin	Char	Dec	Hex	Bin	Char	Dec	Hex	Bin	Char	Dec	Hex	Bin	Char
32	20	00100000	**(Space)**	56	38	00111000	**8**	80	50	01010000	**P**	104	68	01101000	**h**
33	21	00100001	**!**	57	39	00111001	**9**	81	51	01010001	**Q**	105	69	01101001	**i**
34	22	00100010	**"**	58	3A	00111010	**:**	82	52	01010010	**R**	106	6A	01101010	**j**
35	23	00100011	**#**	59	3B	00111011	**;**	83	53	01010011	**S**	107	6B	01101011	**k**
36	24	00100100	**$**	60	3C	00111100	**<**	84	54	01010100	**T**	108	6C	01101100	**l**
37	25	00100101	**%**	61	3D	00111101	**=**	85	55	01010101	**U**	109	6D	01101101	**m**
38	26	00100110	**&**	62	3E	00111110	**>**	86	56	01010110	**V**	110	6E	01101110	**n**
39	27	00100111	**'**	63	3F	00111111	**?**	87	57	01010111	**W**	111	6F	01101111	**o**
40	28	00101000	**(**	64	40	01000000	**@**	88	58	01011000	**X**	112	70	01110000	**p**

Dec	Hex	Bin	Char	Dec	Hex	Bin	Char	Dec	Hex	Bin	Char	Dec	Hex	Bin	Char
41	29	00101001	)	65	41	01000001	A	89	59	01011001	Y	113	71	01110001	q
42	2A	00101010	*	66	42	01000010	B	90	5A	01011010	Z	114	72	01110010	r
43	2B	00101011	+	67	43	01000011	C	91	5B	01011011	[	115	73	01110011	s
44	2C	00101100	,	68	44	01000100	D	92	5C	01011100	\	116	74	01110100	t
45	2D	00101101	-	69	45	01000101	E	93	5D	01011101	]	117	75	01110101	u
46	2E	00101110	.	70	46	01000110	F	94	5E	01011110	^	118	76	01110110	v
47	2F	00101111	/	71	47	01000111	G	95	5F	01011111	_	119	77	01110111	w
48	30	00110000	0	72	48	01001000	H	96	60	01100000	`	120	78	01111000	x
49	31	00110001	1	73	49	01001001	I	97	61	01100001	a	121	79	01111001	y
50	32	00110010	2	74	4A	01001010	J	98	62	01100010	b	122	7A	01111010	z
51	33	00110011	3	75	4B	01001011	K	99	63	01100011	c	123	7B	01111011	{
52	34	00110100	4	76	4C	01001100	L	100	64	01100100	d	124	7C	01111100	|
53	35	00110101	5	77	4D	01001101	M	101	65	01100101	e	125	7D	01111101	}
54	36	00110110	6	78	4E	01001110	N	102	66	01100110	f	126	7E	01111110	~
55	37	00110111	7	79	4F	01001111	O	103	67	01100111	g				